The Schematics have been analyzed from
a die picture by John McMaster.
The chip is identified as:

Fujitsu
672172
C 1985

It is a C1200AVB, MB672172 device
from the AVB-CMOS series.
- 1245 Basic Cells
- 15 Columns, 83 Rows

Cells used (more complex ones):

FJ4 - Power J-K FF with Clear

FJ5 - Power J-K FF with Clear and Preset

FDO - DFF with Reset

FDN - DFF with Set

C41 - 4-bit Binary Asynchronous Counter

C43 - 4-bit Binary Synchronous Up Counter

V1N - Inverter Buffer

K2B - Power Clock Buffer

BD3 - Buffer (Delay Cell)

Delay 11.8ns

V2B - Power inverter

O4T - Output Buffer (Hi-Z)

Not sure this is
the correct type.

007782

Clocks and Reset

File: clocks_and_reset.kicad_sch

Horizontal Signals

File: horizontal_signals.kicad_sch

Vertical Signals

File: vertical_signals.kicad_sch

DMA

File: dma.kicad_sch

DB0 is connected externally to D0 of the data bus

Sync and Blanking

File: sync_and_blanking.kicad_sch

Data Acknowledge

File: data_acknowledge.kicad_sch

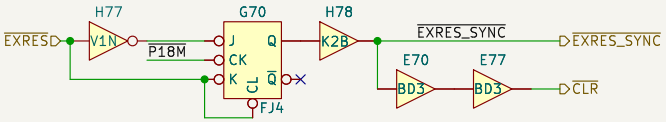
Ulf Skutnabba, twitter: @skutis77

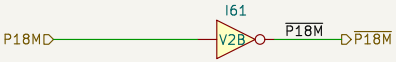
Sheet: /
File: 007782.kicad_sch

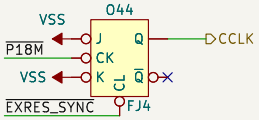
Title: Konami 007782

Size: A4	Date: 2023-08-09	Rev:
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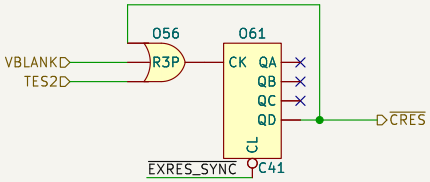
External reset synchronized to P18M clock.







The frequency of CCLK is the
frequency of P18M divided by 2.
f_CCLK = f_P18M / 2 = 18.432Mhz / 2 = 9.216MHz.

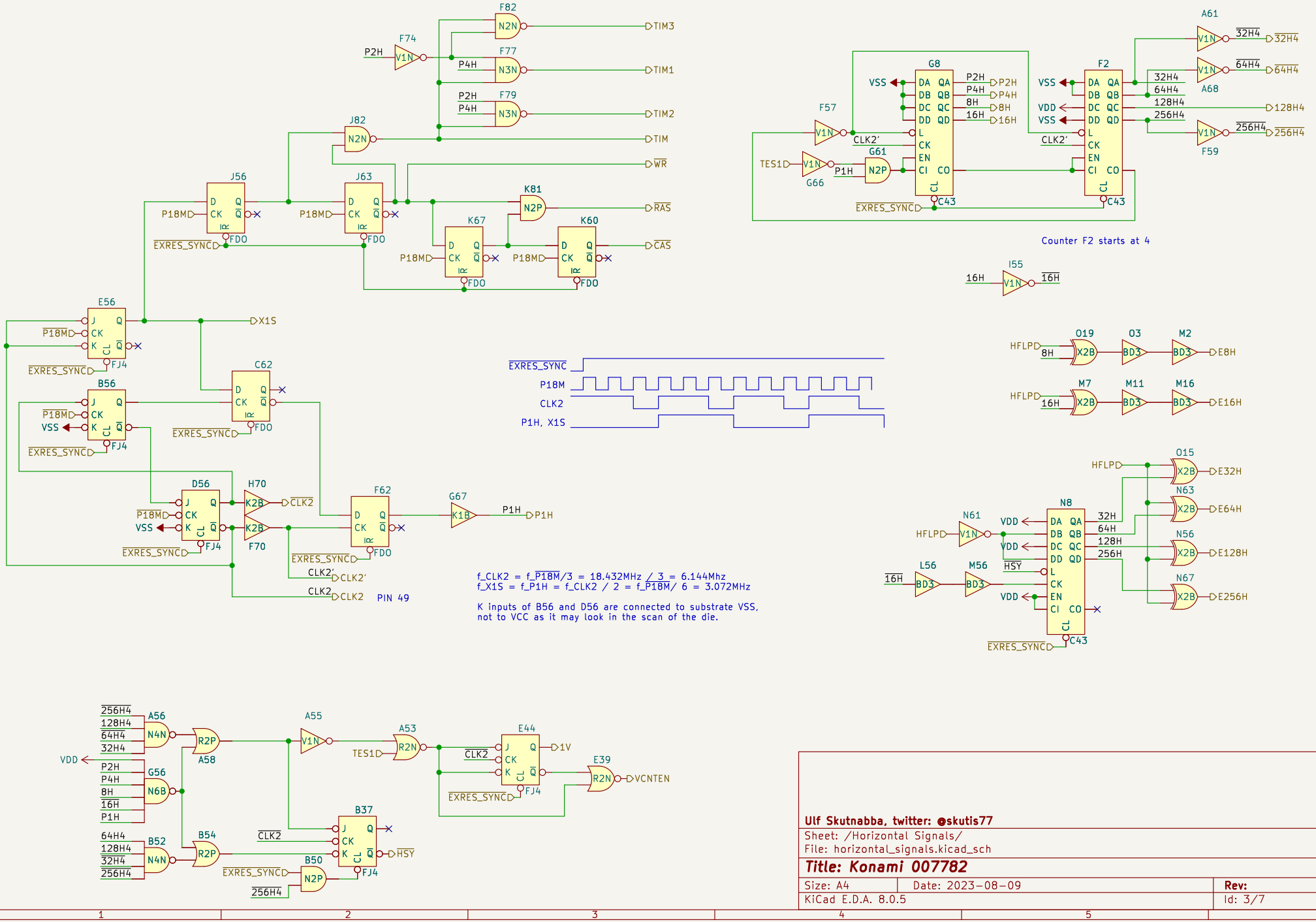


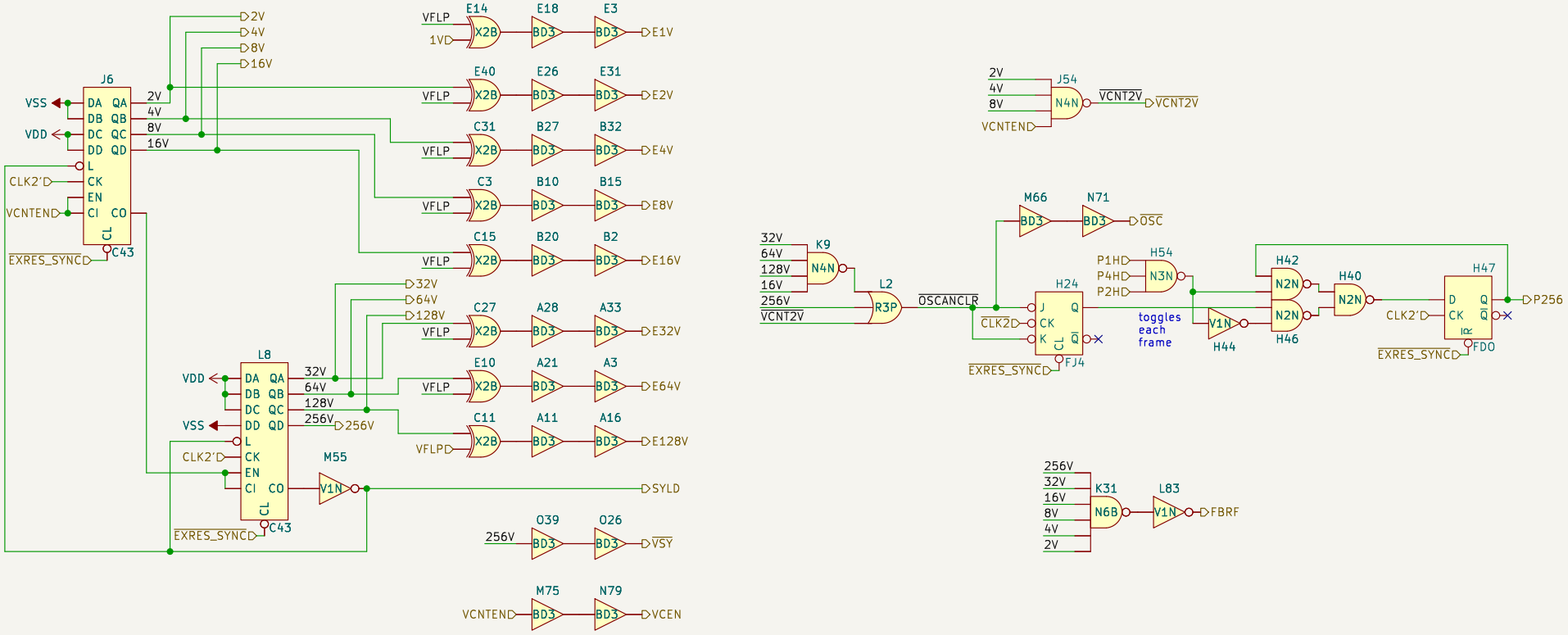
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Sheet: /Clocks and Reset/
File: clocks_and_reset.kicad_sch

Title: Konami 007782

Size: A4	Date: 2023-08-09	**Rev:**
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F82
N2N
TIM3
F74
P2H
V1N
F77
P4H
N3N
TIM1
P2H
P4H
F79
N3N
TIM2
J82
N2N
TIM
WR
J56
J63
K81
N2P
RAS
P18M
EXRES_SYNC
FDO
K67
K60
CAS
E56
X1S
FJ4
C62
B56
VSS
D56
H70
K2B
CLK2
F70
F62
G67
K1B
P1H
CLK2'
PIN 49
f_CLK2 = f_P18M/3 = 18.432MHz / 3 = 6.144Mhz
f_X1S = f_P1H = f_CLK2 / 2 = f_P18M/ 6 = 3.072MHz
K inputs of B56 and D56 are connected to substrate VSS,
not to VCC as it may look in the scan of the die.
G8
F2
A61
A68
F59
F57
G61
G66
TES1
VDD
32H4
64H4
128H4
256H4
C43
Counter F2 starts at 4
I55
16H
HFLPD
O19
O3
M2
M7
M11
M16
E8H
E16H
O15
N63
N56
N67
E32H
E64H
E128H
E256H
N8
N61
L56
M56
HSY
A56
A58
G56
B52
B54
A55
A53
TES1
B37
B50
E44
E39
VCNTEN
1V
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Sheet: /Horizontal Signals/
File: horizontal_signals.kicad_sch
Title: Konami 007782
Size: A4
Date: 2023-08-09
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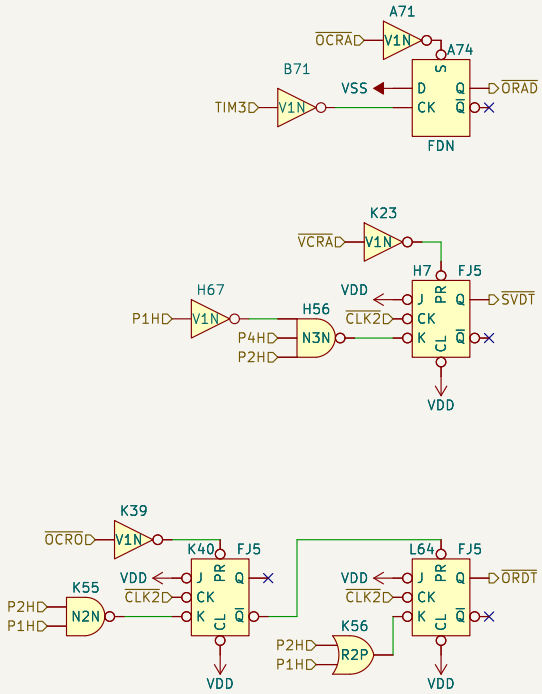


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Sheet: /Vertical Signals/
File: vertical_signals.kicad_sch

Title: Konami 007782

Size: A4	Date: 2023-08-09	**Rev:**
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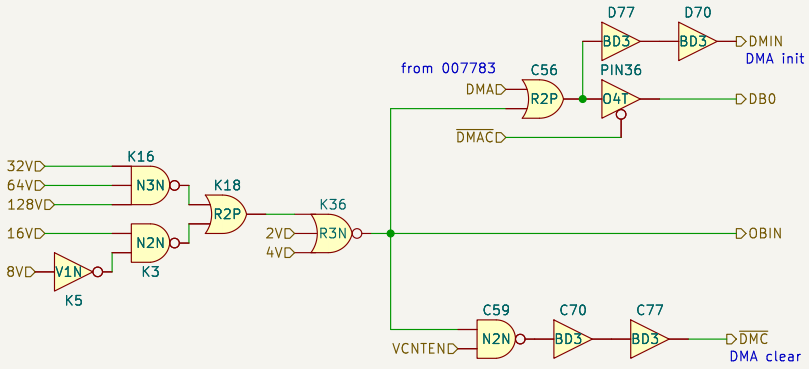


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Sheet: /Data Acknowledge/
File: data_acknowledge.kicad_sch

Title: Konami 007782

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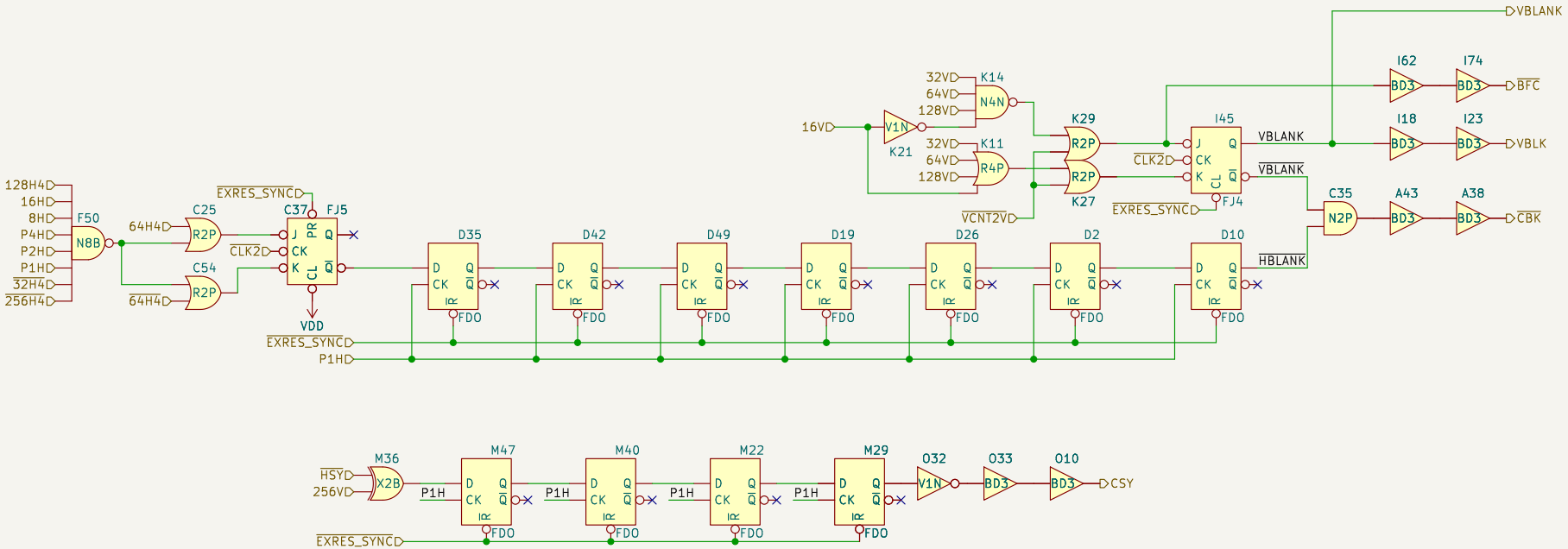


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Sheet: /DMA/
File: dma.kicad_sch

Title: Konami 007782

Size: A4	Date: 2023-08-09	**Rev:**
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Sheet: /Sync and Blanking/
File: sync_and_blanking.kicad_sch

Title: Konami 007782

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